

Introduction to Machine Learning (CS642)

Lecture 01 - Image Classification

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Image classification

- For computers, interpreting the contents of an image is less trivial
- all our computer sees is a big matrix of numbers. It has no idea regarding the thoughts, knowledge, or meaning the image is trying to convey.

Image classification

- In order to understand the contents of an image, we must apply **image classification**, which is the task of using computer vision and machine learning algorithms to extract meaning from an image.
- This action could be as simple as assigning a **label** to what the image contains, or as advanced as **interpreting** the contents of an image and returning a human-readable sentence.
- **Image classification** and **image understanding** are currently (**and will continue to be**) the most popular sub-field of computer vision for the next ten years.

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Image classification

- Image classification is the task of assigning a label to an image from a predefined set of categories.
- let's assume that our set of possible categories includes:
 $\text{categories} = \{\text{cat}, \text{dog}, \text{panda}\}$



Figure: The goal of an image classification system is to take an input image and assign a label based on a pre-defined set of categories.

- More formally, given our input image of $W \times H$ pixels with three channels, Red, Green, and Blue, respectively, our goal is to take the $W \times H \times 3 = N$ pixel image and figure out how to correctly classify the contents of the image.

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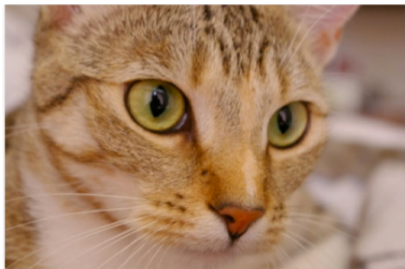
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Image classification

Terminology

- When performing machine learning and deep learning, we have a **dataset** we are trying to extract knowledge from.
- Each example/item in the dataset (whether it be image data, text data, audio data, etc.) is a **data point**. A dataset is therefore a collection of data points
 - ① In the context of image classification, our dataset is a collection of images.
 - ② Each image is, therefore, a data point.

Image classification



151	121	1	93	165	204	14	214	28	235
62	67	17	234	27	1	221	37	189	141
20	168	155	113	178	228	25	130	139	221
236	136	158	230	10	5	165	17	30	155
174	148	93	70	95	106	151	10	160	214
103	126	58	16	138	136	98	202	42	233
235	103	52	37	94	104	173	86	223	113
212	15	179	139	48	232	194	46	174	37
119	81	241	172	95	170	29	210	22	194
129	19	33	253	229	5	152	233	52	44
88	200	194	185	140	200	223	190	164	102
113	16	220	215	143	104	247	29	97	203
9	210	102	246	75	9	158	104	184	129
124	52	76	148	249	107	65	216	187	181
6	251	52	208	46	65	185	38	77	240
150	194	28	206	148	197	208	28	74	93
33	183	248	153	168	205	146	100	254	218
130	53	128	212	61	226	201	110	140	183
165	246	22	102	151	213	40	138	8	93
152	251	101	230	23	162	70	238	75	24
187	105	152	83	167	98	125	180	136	121
139	197	55	209	28	124	208	208	104	40
123	19	144	223	62	253	202	108	47	242
220	144	31	16	136	123	227	62	183	163

29	142	142	75	22	109	111	28	6	5
137	168	41	206	100	70	219	127	114	191
205	154	226	14	89	86	242	67	203	15
247	47	128	123	253	229	181	251	232	28
68	75	24	99	93	63	215	222	102	180
206	246	85	103	215	3	62	64	77	216
126	80	165	149	196	75	186	60	179	193
44	253	164	253	14	216	175	30	46	254
137	23	33	203	241	21	144	63	244	188
32	214	142	121	249	109	99	232	183	71
45	36	152	27	190	137	61	1	237	247
1	14	241	70	2	30	151	67	169	205
32	80	102	32	99	169	91	166	73	214
186	219	9	203	209	240	40	249	119	122
177	252	38	203	119	0	217	139	139	157
154	145	49	251	150	185	235	23	230	156
157	168	223	60	247	118	5	180	16	206
102	208	195	246	140	138	54	191	139	79
17	233	85	169	166	24	49	40	160	97
84	242	247	144	203	3	19	24	198	88
67	67	185	98	123	106	168	105	127	153
37	113	214	252	203	80	146	211	7	16
142	241	66	86	214	133	146	253	189	200
67	215	174	111	189	54	144	56	59	163



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Image classification

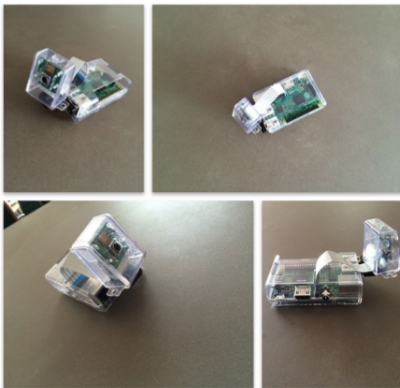
Challenges

- Viewpoint variation
- Scale variation
- Deformation
- Occlusion
- Illumination
- Background clutter
- Intra class variation

Image classification

Challenges

Viewpoint Variation



Scale Variation



Deformation



Image classification

Challenges

Occlusion Variation



Background Clutter



Illumination Variation



Intra-class Variation



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Deep learning classification pipeline

- The four steps of building a deep learning-based **image classifier** as well as compare and contrast traditional **feature-based** machine learning versus **end-to-end** deep learning.

Deep learning classification pipeline

A shift in mindset

```
>>> def fib(n):  
...   if n == 0:  
...     return 0  
...   elif n == 1:  
...     return 1  
...   else:  
...     return fib(n-1) + fib(n-2)  
... >>>
```

Deep learning classification pipeline

A shift in mindset

- Accepts an input, returns an output.
- The process is well defined.
- The output is easily verifiable for correctness.
- Lends itself well to code coverage and test suites.

Unfortunately, this is not the case for deep learning and image classification!

Deep learning classification pipeline

A shift in mindset

- Now, imagine trying to write a **procedural function** that can not only tell the difference between these two photos of cat and dog.
- How would you go about accomplishing this task? Would you check individual pixel values at various (x, y) -coordinates?
- Write hundreds of if/else statements? And how would you maintain and verify the correctness of such a massive rule-based system?
- The short answer is: **you don't**.

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Deep learning classification pipeline

Step #1: Gather Your Dataset

- If we have twice the number of cat images than dog images, and five times the number of panda images than cat images, then our classifier will become naturally **biased** to **overfitting** into these heavily-represented categories.
- **Class imbalance** is a common problem in machine learning and there exist a number of ways to overcome it.

Deep learning classification pipeline

Step #1: Gather Your Dataset (cont.)

- If we have twice the number of cat images than dog images, and five times the number of panda images than cat images, then our classifier will become naturally **biased** to **overfitting** into these heavily-represented categories.
- **Class imbalance** is a common problem in machine learning and there exist a number of ways to overcome it.

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Deep learning classification pipeline

Step #2: Split your dataset

- Now that we have our initial dataset, we need to split it into two parts:
 - ① A training set
 - ② A testing set
- A training set is used by our classifier to “learn” what each category looks like by making predictions on the input data and then correct itself when predictions are wrong.
- After the classifier has been trained, we can evaluate the performing on a **testing set**.

Deep learning classification pipeline

Step #2: Split your dataset (Precaution)

- It's extremely important that the training set and testing set are independent of each other and do not overlap! If you use your testing set as part of your training data, then your classifier has an unfair advantage since it has already seen the testing examples before and “learned” from them. Instead, you must keep this testing set entirely separate from your training process and use it only to evaluate your network.

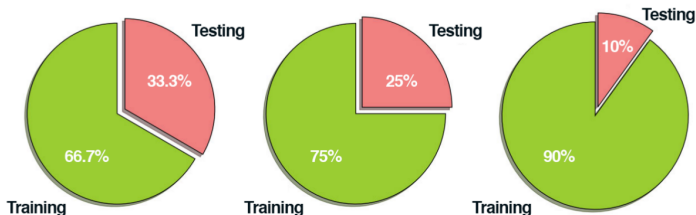


Figure: Examples of common training and testing data splits.

Deep learning classification pipeline

Step #2: Split your dataset (cont.)

- These data splits make sense, but what if you have **parameters** to tune?
- Neural networks have a number of knobs and levers (ex., **learning rate**, **decay**, **regularization**, etc.) that need to be tuned and dialed to obtain optimal performance. We'll call these types of parameters **hyperparameters**, and it's critical that they get set properly.

Deep learning classification pipeline

Step #2: Split your dataset (cont.)

- In practice, we need to test a bunch of these **hyperparameters** and identify the set of parameters that works the best.
- You might be **tempted** to use your testing data to **tweak** these values, **but again, this is a major no-no!** The test set is only used in evaluating the performance of your network.

Deep learning classification pipeline

Step #2: Split your dataset (cont.)

- **Validation set:** Instead, you should create a third data split called the **validation set**. This set of the data (normally) comes from the training data and is used as “fake test data” so we can tune our hyperparameters. Only after have we determined the hyperparameter values using the validation set do we move on to collecting final accuracy results in the testing data.
- **We normally allocate roughly 10-20% of the training data for validation.** If splitting your data into chunks sounds complicated, it's actually not. As we'll see in our next chapter, it's quite simple and can be accomplished with only a single line of code thanks to the **scikit-learn** library.

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Deep learning classification pipeline

Step 3: Train your network.

- Given our training set of images, we can now train our network. The goal here is for our network to learn how to recognize each of the categories in our labeled data. When the model makes a mistake, it learns from this mistake and improves itself.

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Deep learning classification pipeline

Step 4: Evaluate.

- Last, we need to evaluate our trained network. For each of the images in our testing set, we present them to the network and ask it to predict what it thinks the label of the image is. We then tabulate the predictions of the model for an image in the testing set.
- Finally, these model **predictions** are compared to the **ground-truth** labels from our testing set. The ground-truth labels represent what the image category actually is. From there, we can compute the number of predictions our classifier got correct and compute aggregate reports such as precision, recall, and f-measure, which are used to quantify the performance of our network as a whole.

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Feature-based Learning versus Deep Learning

- In the traditional, feature-based approach to image classification, there is actually a step inserted between Step #2 and Step #3 – this step is **feature extraction**. During this phase, we apply hand-engineered algorithms such as HOG [32], LBPs [21], etc. to quantify the contents of an image based on a particular component of the image we want to encode (i.e., shape, color, texture). Given these features, we then proceed to train our classifier and evaluate it.
- When building Convolutional Neural Networks, we can actually skip the feature extraction step. The reason for this is because CNNs are end-to-end models. We present the raw input data (pixels) to the network. The network then learns filters inside its hidden layers that can be used to discriminate amongst object classes. The output of the network is then a probability distribution over class labels.

Feature-based Learning versus Deep Learning

- One of the exciting aspects of using CNNs is that we no longer need to fuss over hand- engineered features – we can let our network learn the features instead. However, this tradeoff does come at a cost. Training CNNs can be a non-trivial process, so be prepared to spend considerable time familiarizing yourself with the experience and running many experiments to determine what does and does not work.

- image classification is the task of assigning a label to an image from a predefined set of categories.

The End